



Evaluating usability for an AR-enabled escape room authoring ecosystem

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ABSTRACT

The increasing ubiquity of mobile devices has enabled the surge of interactive experiences at cultural heritage sites, enriching visitor immersion. However, adapting interactive resources for different settings and types of creators remains a challenge. This paper introduces ENIGMACHINE EDITOR, a web-based authoring tool for crafting augmented reality-enabled adventures inspired by treasure hunts and escape rooms. Designed for curators and hobbyists, it simplifies the generation of custom experiences without prior technical expertise, streamlining development compared to using traditional game engines directly.

The main objective of this work is to comprehensively evaluate the usability of ENIGMACHINE EDITOR for non-technical users, alongside its ability to generate engaging game experiences. A game experience study was first conducted in which participants interacted with games from the platform and subsequently completed the Game Experience Questionnaire (GEQ). Next, two usability studies were conducted in which participants first completed a tutorial with Single Ease Questions (SEQ) after each exercise, and then a series of both short- and long-term tasks on the tool concluded with the System Usability Scale (SUS) questionnaire. The results obtained demonstrate that ENIGMACHINE EDITOR is able to generate satisfactory experiences for museum visitors and highlight desirable usability properties, especially in the short-term test group.

1. Introduction

The ubiquity of mobile devices has allowed the proliferation of applications with which users, using their own devices, can participate in interactive experiences on a wide variety of topics. One context in which this is particularly true is that of museums and other cultural heritage sites. Owing to this, visitors can now easily receive enriched and personalized information along their tours.

In recent years, we have developed several applications of this type, turning the visitor experience into real *treasure hunts* or *escape rooms* with a strong augmented reality component, which we collectively refer to here as *adventures*. To do this, we have made use of tools commonly used in the video game industry, which facilitates development by avoiding the need to create a technological base that would demand a considerable time investment.

Game engines such as Unity,¹ Unreal,² or Godot³ are in themselves complete authoring tools, providing programmers, designers, and artists with truly integrated development environments with which

to develop their games. Even so, they remain generalist tools, and it is often the job of programmers to create specific game elements that designers and artists can then arrange and configure to build the gameplay they are aiming for.

Unity has long been our tool of choice for the creation of all our applications. While the details are not of major importance, in order to separate the technical part, in the hands of programmers, from the creative effort of crafting *adventures*, in the hands of designers, we rely on the use of *scriptable objects*. These are the mechanisms by which Unity allows programmers to define new types of *assets*, beyond maps, textures, models, or sounds, which designers can then manipulate to shape the entire interactive experience.

Despite being a highly valuable tool, the Unity editor is not particularly user-friendly when the number of these *scriptable objects* grows and the relationships between them skyrocket. Additionally, whilst still being conceptually *assets* from an execution point of view, *scriptable objects* are often very close to the way in which programmers view the

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¹ <https://unity.com/>

² <https://www.unrealengine.com/>

³ <https://godotengine.org/>

game and can often cause a wealth of superfluous details to become visible. When designers devise an interactive experience such as a *treasure hunt* or *escape room*, their approach lies at a much higher level, and having to specify it through a myriad of small, interrelated, *scriptable objects* in an environment that is not tailored for it makes their task much more daunting. In addition, this forces them to have a Unity installation, as well as to adapt to and become minimally proficient in an environment that provides a vast battery of functionality. All of this generates a non-negligible amount of noise for the much more restricted work they are required to undertake.

Lastly, in our context, this way of using Unity for the creation of these kind of experiences introduced an additional logistic problem. Over time, we have created many versions of our different adventures to adjust them to specific museum initiatives (e.g., science weeks, nights at the museum, school visits) or to temporary changes in the exhibitions due to artefact loans or restorations of pieces. Most of the core foundations of our games (i.e., programming and art) are common to all of them and are collectively known as ENIGMACHINE. With an increasing number of adventures (created using different *scriptable objects* configurations) performing version control of so many slightly different applications quickly became a struggle.

As a way to address all these problems, in the last few years we have worked on the development of an authoring tool external to Unity, which we refer to as ENIGMACHINE EDITOR, that facilitates the creation of these interactive experiences. This is essentially a web application in which designers can write and specify an entire adventure, providing a clear, distraction-free interface directly within the browser so that it can be used from any device. Each adventure is managed in a distinct project within the application and follows a lifecycle completely independent of the Unity project that will later run the experience. The web application also provides an *export* mechanism through which it creates the *scriptable objects* that ENIGMACHINE expects and thanks to which the final mobile application can be created automatically.

Aside from significantly improving workflow, facilitating version control, and avoiding the need for designers to deal with Unity, the time to create each experience has also been greatly reduced. An informal analysis with designers who were familiar with the nature and inner workings of the experiences created determined that, once an adventure is specified, deploying it to ENIGMACHINE went from requiring a few days to mere hours in most cases. This is especially valuable when taking into account that a substantial portion of the testing and quality control process of these applications must be carried out on-site, outside the development space, with the consequent logistical limitations that this entails.

To confirm that the new authoring tool is also sufficiently accessible to novice users, in this paper we propose a study to test and validate the following preliminary hypotheses:

- **H1.** ENIGMACHINE enables the creation of adventures that offer a rich gameplay experience, in which players perceive high levels of competence, fluency, and satisfaction, and low overall stress.
- **H2.** ENIGMACHINE EDITOR is generally suitable for non-technical users wishing to build adventures from a previous design concept.
- **H3.** ENIGMACHINE EDITOR is also generally useable by non-technical users when designing adventures directly from the platform itself over an extended period of time.

The rest of the paper runs as follows. Section 2 introduces related work, including relevant literature both for gaming applications for cultural heritage and authoring tools in general interactive experiences. Section 3 presents a high-level overview of the ENIGMACHINE system and the AR-enabled games that can be created with it. Section 4 details the user experiments conducted with the aim of validating hypothesis H1 concerning perceived game experience in adventures created with our tool, with Section 5 presents an in-depth explanation of the different usability evaluations of ENIGMACHINE EDITOR with preliminary groups

of design students at our institution to test hypotheses H2 and H3 concerning short and long-term use of the system, with Section 6 outlining the main results of these sessions. Section 7 then summarizes the major findings in the study for each of the proposed hypotheses and concludes the paper outlining directions for future work.

2. Related work

2.1. Gaming applications for cultural heritage

Mobile technology has become a critical component of interactive initiatives for cultural sites, integrating effectively with other approaches, such as the physical objects and artefacts found in museums and exhibitions [1]. Additionally, there is a growing emphasis on “gamification” as a method to increase the appeal of cultural material to learners, as showcased in recent literature reviews [2–6].

The widespread adoption of AR and VR technology is noteworthy in this regard. According to Kahn et al. [5], 46.66% of the 45 publications assessing gamification strategies between 2015 and 2020 propose the use of AR or VR components to explore cultural heritage. Despite often being regarded as more captivating, serious games pose an array of difficulties for creators in that they are required to reach a careful balance between enjoyment and education [7]. While several general guidelines and heuristics exist for educators, researchers, instructional designers, and developers [6], crafting games that guarantee meaningful learning and present a genuine and respectful image of culture remains a challenging problem [2].

The notion of design patterns was first presented by Christopher Alexander in the context of architecture [8] and later adopted in the field of computer science by Gamma et al. [9]. Formalized in the game design domain by Björk et al. [10] in the mid-2000s, design patterns have recently found applications in AR-driven experiences for museums and exhibitions [11]. Namely, Linda et al. [11] focus on patterns that contain knowledge and functions for interactions like spatial connections or general overviews. These patterns enable content creators to focus on templates that learners want to adapt for particular exhibits rather than on devising elaborate templates from scratch. This strategy is consistent with our methods at ENIGMACHINE, where we design educational escape rooms using escape patterns and templates.

Based on the treasure hunt concept, escape rooms target players wishing to seek clues, solve puzzles, and complete certain tasks within a particular period of time, usually with the goal of escaping the room [12,13]. In the recent past, this genre of entertainment has gained immense popularity in major cities and has begun to gain traction in academia [14]. Furthermore, the use of escape rooms for educational purposes has become increasingly popular across different levels of education, including higher education institutions [15]. When properly designed, escape rooms can provide an engaging environment that motivates learners to complete the tasks at hand [16] and also can provide a way to facilitate learning through failure [17]. ENIGMACHINE has also adopted this format, progressively moving the focus away from simple treasure hunts [7] to allowing for the design of rich, fully fledged educational escape rooms where learners solve tasks in more holistic contexts.

2.2. Authoring tools for interactive experiences

Despite the potential of these technologies and concepts, authoring remains a significant challenge in the medium to this day [18,19]. The technical skills required to develop games for both cultural heritage and educational institutions can act as a major barrier to entry, preventing creatives or educators from accessing the field, while the production process can prove prohibitively expensive for institutions that do not possess the financial resources for it. The accessibility of the medium can thus be considered a major challenge for this field, especially given the already existing concerns and difficulties in the adoption of new

technologies, both in the cultural heritage [18,20] and educational domains [21].

The collective solution adopted by the research community appears to be focused on the development of authoring tools that enable users such as educators or curators, who are generally non-technical, to design interactive activities and narratives and translate them onto digital platforms. These efforts fall within the research line of what is known as end-user development, i.e., the creation of authoring tools aimed at professionals who may not be technically savvy enough to operate complex systems during the content authoring process [22]. Included here are tools geared towards the creation of augmented reality mobile applications and interactive stories in cultural heritage [23,24], multiplatform guides and experiences for museums [25–27], or narratives with GPS-based interaction mechanisms in urban environments [28].

Recent work about educational escape rooms [29,30] mention the use of commercial authoring tools for creating digital educational escape rooms (DEER), which use digital environments and may be played by participants not present on site, such as Genially [31], Telescope Live [32], Breakout EDU [33], Room Escape Maker [34], and even open-source platforms such as Escapp [35]. While these tools are suitable for remote or hybrid learning environments, they typically lack features tailored to physical gameplay or augmented reality (AR) integration. For example, Genially offers strong support for interactive visual storytelling but lacks spatial interaction. Telescope Live supports real-time collaboration but is primarily designed for digital replicas of physical escape rooms. Escapp, although open source, does not natively support AR features or physical integrations. In fact, at the time of writing, we are not aware of any authoring tool specifically designed for creating physical or AR-supported educational escape rooms such as our ENIGMACHINE ecosystem.

These efforts do, however, demand careful attention to interface design and user experience across individual authoring tools. While numerous works and heuristics exist that focus on deriving design principles and evaluations for the games or interactive experiences themselves [36–41], whether for commercial games, serious games, or other educational or cultural heritage activities, a comparatively small fraction of this work addresses a similar analysis of the actual authoring tools. A number of state-of-the-art reviews on the issue explicitly highlight this limitation [42,43], stressing how the current spotlight lies primarily on the audience space as opposed to that of the authors, resulting in a landscape where few guidelines or studies exist examining what constitutes an effective end-user authoring tool. This scarcity could potentially stem from the complex nature of this kind of analysis, which often requires long experimental user sessions in order to meaningfully probe the characteristics of the system [44].

3. Description of the ENIGMACHINE platform and games

As stated previously, the system presented in this paper, ENIGMACHINE, constitutes a platform aimed at facilitating the construction of mobile games with significant augmented reality components. These games are built on the basis of the concepts of treasure hunts and escape rooms, and in what follows will be collectively referred to as *adventures*. The main objective of these adventures is to enhance visitors' immersion and involvement in museums and other similar spaces while encouraging interaction with the various elements of the exhibitions in which they take place.

This system is comprised of a web editor for content creation, known as ENIGMACHINE EDITOR, and a mobile application to distribute the experience resulting from the design process within the tool. Games generated with ENIGMACHINE feature a narrative that places the player as the protagonist of the adventure, as they navigate a series of interconnected puzzles, riddles, and quests interwoven into the main story. Structurally, adventures are divided into stages or phases from among the 13 types available at the time of writing. These phases,

which can be interpreted as scriptable authoring templates (i.e., customizable content blueprints that follow a set of preconfigured patterns to dynamically generate or modify content), are grouped into distinct categories based on their underlying design intent (e.g., finding an artefact in the exhibit, collecting responses from participants, posing a puzzle or challenge to be solved by inspecting the environment, etc.), and are summarized in Table 1. These are presented to the player sequentially, requiring the completion of a stage to advance to the next, with the only exception being the Room-type phase, in which a set of parallel sub-blocks of sequential phases are displayed to be solved in any desired order. An essential aspect aimed to be attained is to provide the sensation of deep immersion, offering players the experience of being actively involved in the site visit instead of being merely passive observers of the venue. In order to improve technological accessibility, these games can be run on the smartphones and tablets of both the organizing institution and the visitors themselves. Players can then interact with puzzles by means of touchscreen features, GPS systems requiring them to reach specific geographic locations as part of the narrative of the experience, and augmented reality modules using the device's camera to scan objects or codes in their vicinity.

From a game designer's point of view, the complete process of authoring and deploying an adventure can be summarized as follows:

1. Content creators log into the web application and, via their user account, create the adventure leveraging the tools provided by the platform.
2. Once completed, the web tool generates a mobile application that can be downloaded by the user.
3. The application is installed on the mobile devices used by visitors.
4. Visitors engage with it.

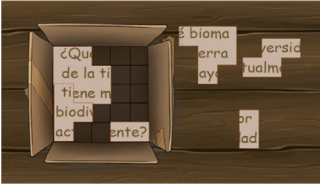
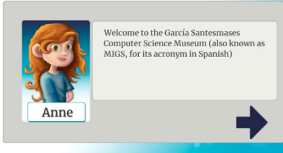
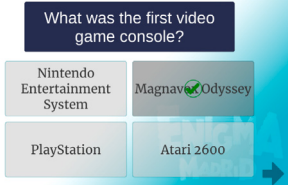
While the model of creating content through the web by non-technical users is not entirely novel and in fact is currently being actively applied by well-known platforms such as *Kahoot!* or *Canva*, ENIGMACHINE EDITOR distinguishes itself by being specifically designed for the creation of escape rooms and treasure hunts with a strong environmental exploration component through AR, QRs and GPS in educational centres or museums, adapting its components to the needs of this context. Following the same approach of some of the mentioned platforms for content creation, our tool organizes the structure of the visit in sequences of building blocks represented as slides.

The web editor is conceived as a single-page solution with React as the main development framework and a supporting server in NodeJS for managing user data, adventures, and metrics. The phase-based modularity described above for the adventures extends naturally to the authoring tool, where each of the stages supported in the game is translated to a dedicated editing component, as shown for selected examples in Fig. 1.

Adventures created using the web editor are serialized to a structured JSON file with the adventure metadata (such as its title, description, characters, and overall settings), followed by the ordered list of stages, each specified with its own parameters according to the stage type. References to multimedia assets, such as character images or AR targets, are stored separately on the server and linked using unique URLs in the JSON. When a content creator completes their adventure, they can request a mobile app to be built through the platform. This triggers a dedicated builder service, which downloads the JSON file and automatically converts it to Unity3D's native scriptable object format, a structure optimized for runtime performance.

The building process can take some time (between 5 and 10 minutes for a typical adventure), however, as it involves assembling and packaging all the contents of the adventure into an application. To accommodate this efficiently, builds are managed by secondary “workers” running in the background. Once a requested build is complete, the frontend is automatically notified by the system to let the user

Table 1
Types of phases in Enigma games.

Phase Types	Description	Example
<i>AR-Scan</i> <i>AR-Overlay</i> <i>QR-Scan</i> <i>Reach GPS Spot</i>	Ask players to explore the venue and scan a given object using their device's camera in different fashions, or reach a certain GPS location in the vicinity.	
<i>Pack Puzzle</i> <i>Number Lock</i> <i>Password</i>	Ask players to solve a puzzle to progress, such as a four-digit code, a password stage, or a tile packing level.	
<i>Narrative</i> <i>Object Obtained</i> <i>Diary Page</i>	Show a narrative phase to players in which characters interact with one another, or the player obtains an item or diary page.	
<i>MC Test</i> <i>MC Free Answer</i>	Get players to choose from a set of multiple choice options either in a quiz fashion or as a free answer.	
<i>Room Module</i>	Have players solve blocks of puzzles in an escape room fashion, collecting clues to help them unlock a final puzzle block.	

know that their app is ready to download. This asynchronous process enables users to continue editing or creating new adventures without idly waiting for builds to finish.

The builder then runs the Unity batch editor to create a standalone mobile application that contains the complete adventure and all the assets needed. The app is therefore completely self-contained and requires no Internet connection once installed. This is in contrast to platforms such as *Kahoot!*, which load content on demand at runtime, while *ENIGMACHINE* pre-assembles everything for robustness and offline operation. In addition, experienced Unity users can also bypass the web editor and directly create or edit adventures within the Unity template project, allowing for deeper customization, such as altering background visual effects or fonts or adding entirely new phase types.

4. Experimentation on gameplay experience

One of the main objectives of the design and development of this work is to enable the creation of augmented reality video games that combine, on the one hand, exploration of the environment and, on the other hand, a satisfying game experience that is motivating and entertaining for users while encouraging discovery and enjoyment of the different elements of the museum. To achieve this, a study was conducted to evaluate game experience using different adventures generated with the tools described in this paper. This evaluation aims to

observe and analyse user feedback in order to identify strengths and weaknesses in the balance between enjoyment and difficulty of the generated games.

In the last decade, game research has become increasingly prominent in human-computer interaction studies, leading to the development of several self-assessment instruments aimed at evaluating gameplay experience. Among others, the Game Experience Questionnaire (GEQ) and its variants [45] have been extensively employed by practitioners across a wide spectrum of genres, interactive experiences and user types [46,47]. Due to this popularity in the literature, the in-game GEQ was used here, administered to the players after completing the adventure assigned to them. Key aspects such as level of immersion, fluency, positive and negative affect, challenge, tension, and anxiety were assessed. Additionally, information was collected on users' previous experience with escape rooms along with basic demographic data.

Regarding the design of the experiment, two variations of an adventure game set on the third floor of the García Santesmases Museum, dedicated to the history of computer science and located at the Faculty of Computer Science of the Complutense University of Madrid, Spain, were constructed. These were then administered to different groups of users working in teams of 2 to 4 players. The adventures designed comprised a total of 8 blocks of puzzles, making use of most of the types of phases available in the authoring tool. Each of these puzzle blocks

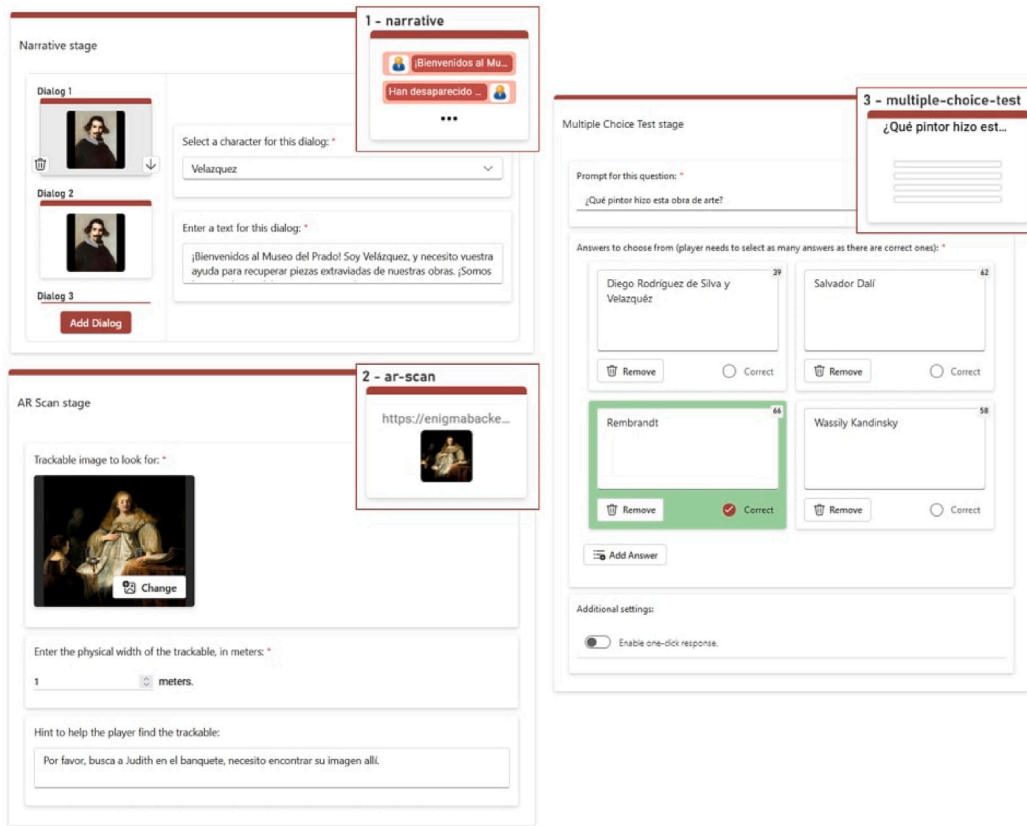


Fig. 1. Sample Editors for Narrative, Multiple Choice, and AR-Scan Stages.

required the careful inspection of at least one artefact or collection of artefacts in the museum and were distributed in two *Room*-type stages, each confined to one of the wings of the area where the activity took place. As mentioned above, the 4 blocks in each room could be completed in any order, being shown to the players in a random arrangement in each game.

The experiment was carried out across multiple sessions with a total of 62 participants, including 17 females, 44 males, and one person who chose not to answer the question. The mean age was 24.58 years (standard deviation = 3.05; minimum age: 19 years, maximum age: 34 years). Among the participants, 61.3% had little or no experience with escape rooms, 29% had moderate experience, and 9.7% considered themselves skilled players. Advertisements targeting our educational institution's design, programming, production, and art students were used to recruit participants, and their involvement was scheduled into one-hour blocks of museum visits.

In each of these sessions, participants were randomly split into teams of 2 to 4 people, each receiving a tablet with the pre-installed game, as well as sheets of paper and writing pens to take notes if they deemed it necessary. Initially, given that the museum can receive several groups of visitors at the same time, the simultaneous participation of multiple teams in the same time slot was allowed in cases where time restrictions required it. However, to avoid overcrowding of players around certain key points in the museum or having groups blindly following others to copy their answers, alternative versions of the game were administered in these sessions, with randomized sections and distinct starting points for each team.

Once the equipment and materials had been prepared, players were free to interact with the game in the corresponding section of the museum, with a one-hour time limit, after which they were instructed to return to their starting point to conclude the activity. These sessions were designed with competition in mind, by announcing that the first players to complete the escape room would receive a small reward

Table 2

GEQ results by dimension (N = 62).

Dimension	Mean	Std	Median
Competence	2.69	0.79	2.75
Sensory and imaginative immersion	2.06	0.72	2.0
Flow	2.12	0.97	2.0
Tension	0.46	0.59	0.5
Challenge	2.36	0.78	2.5
Negative affect	0.29	0.46	0.0
Positive affect	3.28	0.66	3.5

in order to encourage immersion and rivalry between teams. After finishing the adventure, users were asked to submit the 14 items of the In-Game GEQ questionnaire [45].

Participants rated each item on a five-point Likert scale, where a value of 0 indicated complete disagreement and a value of 4 indicated complete agreement. The GEQ components of the game are grouped into pairs of items categorized as follows: Competence (items 2 and 9), Sensory and Imaginative Immersion (items 1 and 4), Flow (items 5 and 10), Tension (items 6 and 8), Challenge (items 12 and 13), Negative Affect (items 3 and 7), and Positive Affect (items 11 and 14). In addition, and on a voluntary basis, each participant had the opportunity to provide an open text commentary outlining what they enjoyed most and least about the experience, as well as their personal qualitative evaluation of the session. The only incentive for participants to take part in the study was the opportunity to try out an escape room with augmented reality elements in the museum, with minor prizes for the first ones to complete it.

4.1. Results

Overall results are reported below following data collection and questionnaire analysis. The distribution of GEQ scores by dimension

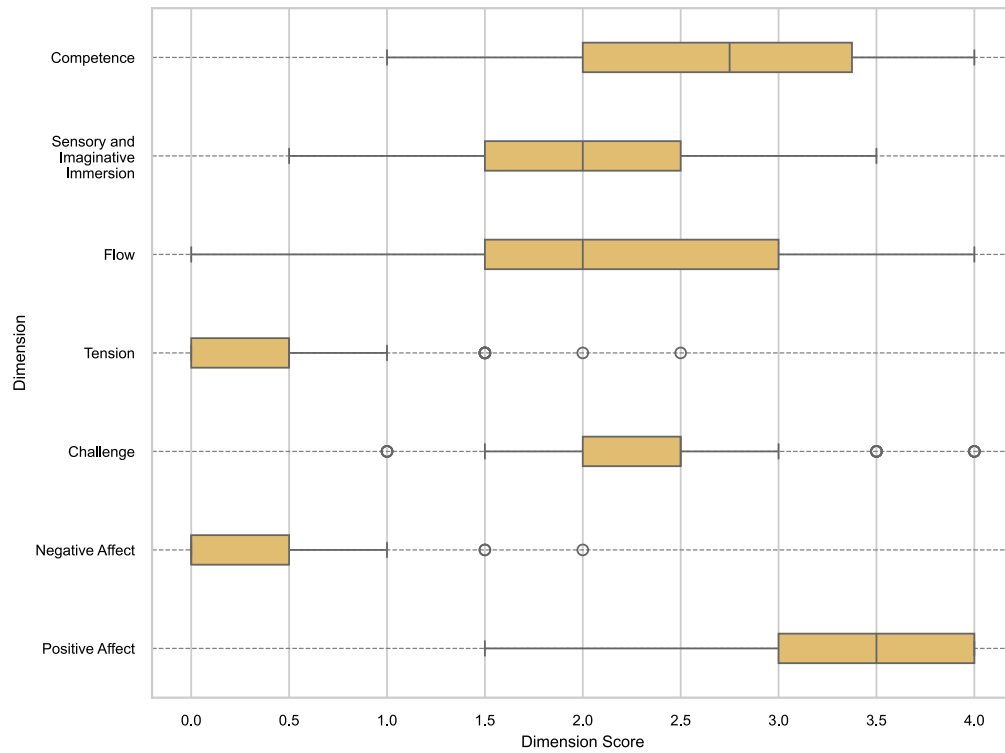


Fig. 2. Distribution of GEQ Scores by Dimension.

is represented visually in Fig. 2, with the aggregates for mean score, median, and standard deviation detailed in Table 2. In general, most players exhibited very low levels of stress and negative affect, with high levels of positive affect with the experience. Perceived levels of challenge were moderate for much of the population, as were levels of competence, flow, and sensory and imaginative immersion.

Of the latter, the results for flow and immersion merit closer inspection, as they report the weakest overall minimum and mean scores (mean for flow = 2.12, mean for immersion = 2.06). If we look at the sub-dimensions that make up these components, we can see that the worst results are found in item 1 (“I was interested in the story of the game”, mean = 1.79) and item 12 (“I felt completely absorbed”, mean = 1.97). Analysing the open-ended user responses, we can trace the origin of these results back to the following key points:

- A number of users note their preference for shorter textual explanations, with a greater focus on puzzles than on narrative. In other instances, players report favouring other multimedia formats such as video or sound over text in order to engage with the story of an escape room.
- Several players found the sharing of the museum space with other teams participating in the experience to be a source of friction, especially when more than one group was simultaneously tasked with solving puzzles located in close proximity to each other in the exhibit.
- Some participants described a degree of friction over usability issues such as not being able to go back in narrative stages to revisit previous messages. Furthermore, a significant number of the concerns regarding immersion were linked to the visual and auditory aspects of the game, which they found unattractive and consequently less immersive than expected.

Other general trends of a more positive nature include:

- The vast majority of participants reported not feeling at all tired or irritable during the experience. Perceived frustration was also very low, albeit marginally higher than the other two negative values as a consequence of the aforementioned points.

- A large number of users reported feeling very good about the experience, with only 5 users expressing feeling less than moderately successful. No participant submitted the minimum score for the item “I felt skilled”.
- No participant answered below “Moderately” for the items “I felt satisfied” or “I felt good”.

Additionally, we sought to analyse whether the perceived game experience varied significantly depending on the users’ prior experience with escape rooms. Of all participants, 61.3% reported having little or no previous experience with escape rooms, with the other 38.7% considering themselves intermediate or advanced players. Given the ordinal qualitative nature of the variables studied, a non-parametric analysis was used to evaluate whether there were significant differences between the variables measured. Due to the samples’ independence, the Mann-Whitney U test [48] was used to compare medians. The only dimension where a statistically significant difference was detected between the groups with and without previous escape room experience ($p < 0.05$) was for the challenge component ($U = 657$, $p = 0.001$, median without experience = 2.5, median with experience = 2).

5. Experimentation on authoring tool usability

Authoring tools are a critical part of the creation and research of interactive experiences, impacting the authorship process and the resulting works. While numerous tools dedicated to different types of multimedia artefacts exist, comparably limited work has been done to evaluate the user experience aspects of these applications relative to the effort invested in the assessment of the actual artefacts generated with these tools. In the field of IDN (interactive digital narrative) authoring tools, for instance, a 2023 literature review [19] revealed that only 7 out of the 37 authoring tools examined had conducted a thorough UX evaluation, with 26 of them having simply introduced the tool along with some examples and demonstrations, but no usability analysis to further validate them with intended end-users.

This same study raises possible hypotheses for the reasons behind this observation, emphasizing the following fundamental points:

- It is challenging to break down a complex creative process for task-based usability tests.
- Content authoring is often a much lengthier process than typical use cases in the systems for which conventional usability analyses are designed, and short-term evaluations are consequently not representative of real tool usage.
- Quantitative metrics are useful but, by themselves, insufficient to truly understand the experience of creatives.

Completing a series of mechanical tasks where a clear goal exists for the user is effectively not directly comparable to dedicating an extended period of time to operating a tool to design multimedia content. This is not to say, however, that traditional usability analyses are of no value, but it does highlight the need for more comprehensive studies that consider the properties of the system both when used to perform mechanical, time-bound, and domain-bound tasks and to develop end-to-end projects.

For these reasons, in this section we discuss the experiments conducted to evaluate the user experience in ENIGMACHINE EDITOR in both settings, starting with a preliminary usability analysis in short-term sessions with bounded tasks and a modest group of participants in Section 5.1 and continuing with a roughly one-month-long evaluation of a creative task in Section 5.2, with a significantly larger participant group. The results of these studies, both quantitative and qualitative, are then reported in detail in Section 6.

5.1. Evaluating usability in design implementation tasks

Our second hypothesis raised the question of whether a user without a technical background or prior familiarity with the tool's concepts is generally able to use ENIGMACHINE EDITOR for what we refer to as "design implementation tasks". By this, we denote tasks in which a user "translates" a set of design ideas into a working prototype, without these being necessarily devised by the user, assuming that everything included in these ideas is feasible to implement with the tools provided. Conceptually, this corresponds to the "short-term" system usage discussed at the beginning of this section. To assess this property, a study was conducted to analyse the ability of non-programmer users to convert a design document into a complete adventure using ENIGMACHINE EDITOR.

This evaluation aims, on the one hand, to observe user feedback to understand the possible frictions and mechanical complexities perceived when attempting to translate a design concept into a functional element on the platform and, on the other hand, to inspect the games created by these users to determine the degree of success achieved in the implementation process. For this purpose, the System Usability Scale (SUS) questionnaire [49], administered to the participants after the end of the testing sessions, was used in tandem with Single Ease Question (SEQ) questionnaires [50] after each task completed by the users in a first tutorial stage at the beginning of the experiment. Both of these evaluation systems have been repeatedly demonstrated to be valid, sensitive and reliable for UX evaluation. Key properties such as ease of use, complexity, consistency, confidence in use, and the learning curve of the system were assessed.

The experiment was designed as a practical session in the context of the Master's Degree in Video Game Design at our educational centre, with the participation of 14 individuals, comprising 5 women and 9 men, with a mean age of 24.88 years (standard deviation = 2.31; minimum age: 22 years, maximum age: 34 years). The participants were university design students, most with little or no programming experience, and with non-technical profiles. All data collected during the session were anonymized prior to analysis.

The session began with a brief introduction aimed at contextualizing the activity and the tool. This was followed by a self-guided tutorial integrated into the web application, consisting of a battery of tasks to introduce the most important sections of ENIGMACHINE EDITOR. This

tutorial was comprised of 13 tasks presented through textual prompts, allowing the user the option to quit any of them if they felt unable to complete it successfully. Once a task was completed, and before proceeding to the next task, the system displayed a single-item SEQ questionnaire with the prompt, "Overall, how difficult or easy was it to perform this task?". Responses to this question were given on a 7-point Likert scale, where a value of 1 indicated great difficulty in completing the task and a value of 7 indicated great ease.

Once the tutorial phase was completed, participants were given a draft design document for the game used in the experiments from Section 4, outlining the structure of the game flow and the different puzzles to be incorporated, referencing the museum artefacts needed to overcome each challenge. In addition, they were provided with a reference adventure, which was blank save for having the necessary assets pre-included to build the game (i.e., images required for AR events or to define characters, for instance). After the material was provided, participants were given 2 h to translate the design document into a working implementation using the authoring tool. Subsequently, and for each participant, the number of essential phases (i.e., ignoring cosmetic or split narrative phases) correctly translated into ENIGMACHINE EDITOR from the design document in the time allotted was computed. We consider that a minimum implementation of the draft script includes at least 18 stages, including those nested in room-type phases.

Finally, and briefly before concluding the session, participants were asked to fill in the SUS questionnaire designed to assess the overall usability perceived by users during the experiment. This questionnaire consists of 10 items to assess components such as ease of use, complexity, consistency, confidence in use, and the learning curve of the system.

Participants rated each item on a 5-point Likert scale, where a value of 1 indicated complete disagreement and a value of 5 indicated complete agreement. In addition to analysing the results for each of the questions independently, the SUS score of each user was computed following the formula proposed in the original SUS manuscript [49]. Additionally, and on a voluntary basis, each participant could write a free text commentary elaborating in more detail how they had felt about the tool, both positively and negatively.

Section 6 enumerates the question sentences included in the survey and details the result of the evaluation.

5.2. Evaluating usability and user experience in creative authoring

The final research issue of this paper raises the question of whether ENIGMACHINE EDITOR is a generally useable tool for non-technical audiences to design interactive experiences directly and from scratch from the web platform itself over a longer period of time. The main difference between this use case and the one from the previous section is that we are now considering situations where the user is not a priori handed a set of design ideas for which a translation is guaranteed to exist in the tool but a much more ambiguous and creative task for which they are responsible for crafting a comprehensive design solely employing the features provided by ENIGMACHINE.

Our hypothesis prior to the start of the experiments was that this type of evaluation would yield overall weaker usability results than those observed in the design implementation tasks, as a consequence of the potential inability to translate certain ideas from the creative process into the palette of features provided by the system, and the issues with traditional usability evaluation assumptions outlined at the beginning of this section. Here we considered hypothetical situations such as users wishing to include conditional flows, re-skinning assets from the final game, or making more complex puzzles than those available in ENIGMACHINE.

This experiment was set up as a practical project in the video game design subject of the video game development undergraduate degree at our educational centre, with the participation of 50 individuals. The participants were first-year undergraduate students, most of whom had

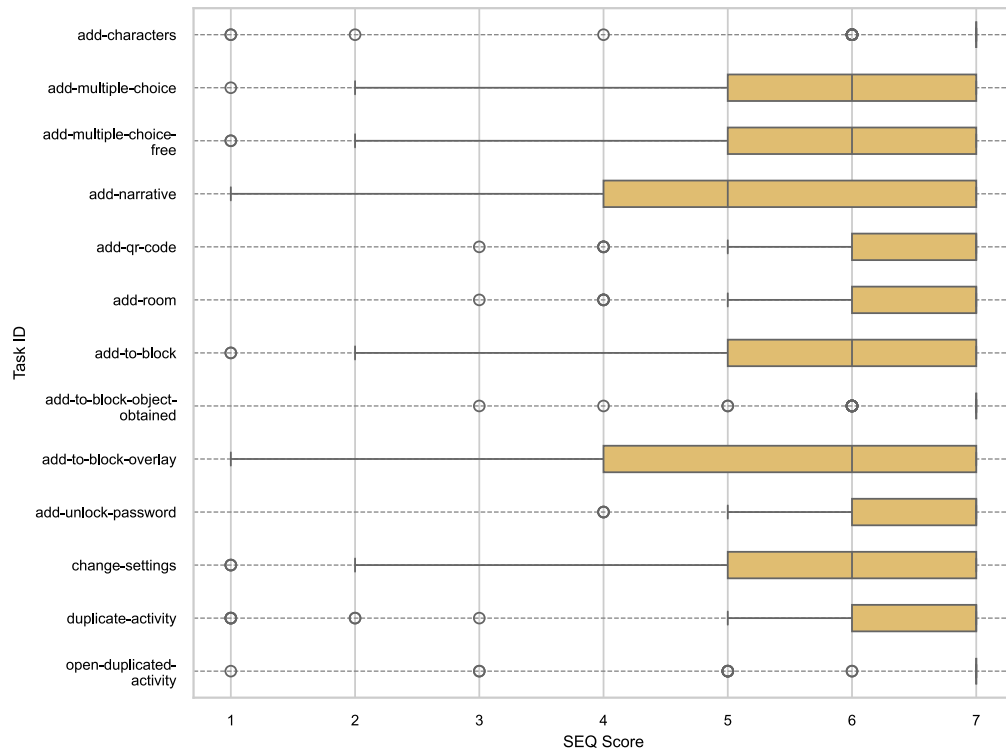


Fig. 3. Distribution of SEQ Scores by Task.

Table 3

Overview of evaluation types conducted with ENIGMACHINE.

Evaluation type	Setting and length	Questionnaires
Game Experience of ENIGMACHINE Games	García Santesmases Museum visitors, one-hour sessions	GEQ, Open-ended questions
Short Term Usability of ENIGMACHINE EDITOR	Design Students (bachelors and masters), three-hour sessions	SEQ, SUS, Calculated Accuracy
Long Term Usability of ENIGMACHINE EDITOR	Design Students (bachelors), one-month free design	SUS, Open-ended questions

little or no programming experience at the time the experiment was carried out.

In this context, the participants, in groups of 3 to 4 students, and over a period of time of roughly one month, were tasked with using the web editor to build their own adventures based on specific areas of the García Santesmases Museum, which was the same as the one used for the experiments in Section 4. During this time, participants were granted access to the museum venue to take photographs and gather information about the different artefacts available in the exhibitions, with the aim of constructing rooms, narratives, and puzzles that would motivate visitors to inspect the museum's objects carefully during the game. In the final stages of construction of these adventures, and on the occasion of a series of visits to the museum in the context of open days for secondary schools, a set of immersive tours was organized using the games created by those students interested in having live beta testers for their games. These were organized in 3 sessions of 37, 45, and 23 visitors, respectively, coming from a total of 6 educational centres in our city and mostly between 15 and 18 years old.

The usability questionnaires employed, as well as the tutorial and tool contextualization sessions in this experiment, were identical to those in Section 5.1. Thus, the initial phase of both groups was the same, with both contributing to the tutorial data with individual tasks linked to SEQ surveys, with the main differences being located in the second phase of the experiment. In this latter case, all participants were administered the SUS questionnaire after the end of the period allotted to complete their adventures.

An overview of the different evaluations conducted with our system can be found in Table 3.

Table 4

SEQ results for tutorial tasks (N = 73).

Task ID	Mean	Std	Completion %
add-characters	6.53	1.20	95.89
add-multiple-choice	5.90	1.38	90.41
add-multiple-choice-free	5.70	1.50	82.19
add-narrative	5.22	1.57	94.52
add-qr-code	6.42	0.94	89.04
add-room	6.25	0.98	87.32
add-to-block	5.72	1.56	85.51
add-to-block-object-obtained	6.67	0.75	94.03
add-to-block-overlay	5.33	1.74	86.76
add-unlock-password	6.49	0.79	93.06
change-settings	5.75	1.53	94.52
duplicate-activity	6.15	1.65	100.0
open-duplicated-activity	6.64	1.06	100.0

6. Results

The results obtained in the usability evaluations on design implementation and creative authoring tasks are presented in this section after data collection and questionnaire analysis.

Starting with the data collected from the tutorials, looking at the joint results for both groups in Table 4 (and a visual distribution in Fig. 3), it is possible to gather detailed insights into the specific tasks that lead to the most conceptual problems. Note that responses are given on a scale of 1 to 7 following the SEQ Likert format. It is noteworthy here that the most challenging assignments in terms of both SEQ and success rate are those related to modifying room-type stages by incorporating and editing new nested blocks within them. This is also to be expected, given that dealing with a nested structure is clearly more complex than working with individual stages. Multiple-choice questions also introduce a certain degree of fiction, especially those with free-response answers. Nonetheless, the vast majority of the tasks exhibit satisfactory results, with no more than a small set of users reporting low SEQ values in any of them.

Table 5
Test results for SUS prompts.

Prompt		Median	U	p-value
1. I think that I would like to use this system frequently.	DES IMP	3.0 3.0	154.0	0.12
2. I found the system unnecessarily complex.	DES IMP	2.0 1.0	342.0	0.01
3. I thought the system was easy to use.	DES IMP	4.0 5.0	104.5	0.01
4. I think that I would need the support of a technical person to be able to use this system.	DES IMP	2.0 1.0	287.0	0.16
5. I found the various functions in this system were well integrated.	DES IMP	3.0 4.0	82.5	0.0
6. I thought there was too much inconsistency in this system.	DES IMP	3.0 1.0	382.5	0.0
7. I would imagine that most people would learn to use this system very quickly.	DES IMP	4.0 4.0	186.5	0.36
8. I found the system very cumbersome to use.	DES IMP	2.0 1.0	305.5	0.07
9. I felt very confident using the system.	DES IMP	3.0 4.0	72.5	0.0
10. I needed to learn a lot of things before I could get going with this system.	DES IMP	2.0 2.0	228.0	0.96

From an overall usability point of view, the SUS Score was computed for each of the users following the SUS Score formula in [49] (which outputs a score between 0 and 100), resulting in a mean of 80.83 (standard deviation = 15.31, median = 82.5) for the design implementation task group, and 63 (standard deviation = 14.86, median = 65) for the creative authoring task group. According to a review including data from 241 industrial usability studies [51], the usability of the system in the first task group would approximately place in the 90 to 95 percentile range (grade A in the study), with the second group placing slightly below the industry average of 68 points (grade C in the study) and would benefit from improvements in the usability of the system. This corresponds to the expectations prior to conducting the experiments, taking into account that the design implementation tasks are clearly simpler and narrower in scope than those of the second test, and also more similar to the typical usability analysis setting focusing on specific tasks instead of wholistic, long-term creative evaluations.

Instead of focusing solely on an overall usability index, the participants' responses to each of the SUS questionnaire items are broken down below to help identify where the system's usability succeeds and fails. In what follows, note that item scores are given on a scale of 1 to 5 following the SUS Likert format. Given the ordinal qualitative nature of the variables under enquiry, a non-parametric test was used to determine whether significant differences existed between the variables examined. Due to sample independence, the Mann–Whitney U test was utilized to compare medians. Table 5 suggests statistically significant differences ($p < 0.05$) for items 2, 3, 5, 6, 8, and 9, but no evidence for this in the other questions. Note that in this table, “IMP” stands for the design implementation group and “DES” for the long-term creative design one. We discuss these trends in greater detail below.

- While significant differences are discernible, neither group consistently found the system unnecessarily complex (question “2. The system was unnecessarily complex”, with means of 1.44 and 2.16 for implementation and creative authoring, respectively), nor very complicated to use (question “8. I found the system very complicated to use”, means of 1.33 and 1.84). In practice, only one participant in both groups rated the latter question with a value above 3.
- While both groups agree on the overall ease of use of the system (question “3. I found the system easy to use”, means of 4.56 and 3.7), those assigned with design implementation tasks found it significantly easier than those with creative authoring ones. As

discussed above, this is fitting in that the second group was given a significantly less constrained assignment with a higher degree of freedom, where no obvious correspondence between design intent and tool feature existed.

- In line with the previous pattern, the implementation group felt significantly more confident than the creative authoring group (question “9. I felt very confident using the system”, means of 4.33 and 3.00), which is explainable in light of the more guided nature of the first experiment.
- Lastly, the creative group perceives a significantly higher level of inconsistency in the system (question “5. I found the various functions of this system to be well integrated”, with means of 4.22 and 2.98, and question “6. There was too much inconsistency in this system”, with means of 1.44 and 2.76). While this point requires further analysis, we hypothesize that extensive use of the tool enables the user to explore it more thoroughly and engage with it closely enough to spot areas involving tedious sequences of steps or usability patterns that are dissonant with either their mental model or with the rest of the application's elements.

The non-significant items yield generally desirable system properties from the users' perspective. The majority of participants do not anticipate that they would require technical support to work with the tool (question “4. I think I would need the support of a technician to be able to use this system”, with means of 1.56 and 1.96), consider the system easy to learn (question “7. I would imagine that most people would learn to use this system very quickly”, means of 4.0 and 3.78), and do not perceive a high entry barrier (question “10. I had to learn a lot of things before I started using this system”, means of 2.22 and 2.14).

Lastly, for the design implementation experiment, the success rate for each of the users was computed as the number of items correctly translated from the design document to the authoring tool in the time provided, divided by the minimum number of steps required to represent the proposed adventure, resulting in a favourable success rate of 85.8%.

On a qualitative level, the open-ended text comments of the participants of the creative authorship experiment can be summarized in the following key points:

- Designers request the inclusion of additional settings to customize the interface of the resulting game and enhance player immersion. In some instances, these suggestions also include comments aimed

at introducing mechanisms to customize the music and sound effects of the generated adventure, as well as the backgrounds, colours, and other visual elements of each stage.

- Another prevalent request is the inclusion of flow control systems that allow for the addition of a notion of “consequences for good and bad decisions” to the adventure. This includes ramifications on the game’s narrative or scoring systems based on player performance.
- Designers working in groups express a demand for tools geared towards collaborative and structured game design. In the current version, they argue that while it is easy to work with ENIGMACHINE EDITOR to implement a design, the platform does not provide suggestions or design ideas in the way that brands such as Kahoot or PowerPoint do, delegating the full weight of the design to the user.

7. Conclusions and future work

This paper presented an authoring tool designed to facilitate the development of AR-enabled adventures in cultural heritage contexts. The main goal is to empower content creators with tools that allow them to generate rich and immersive experiences, minimizing friction with the system and providing designers with an authoring process that is as useable and accessible as possible.

The three hypotheses underpinning this study concerned the game experience of the adventures generated with the ENIGMACHINE system and the usability of the tool for users with non-technical profiles when creating experiences with and without prior design concepts. The results obtained provide information about the quality of our ecosystem in these three dimensions as perceived by users and players.

Hypothesis H1 was confirmed, as the adventures built with the system were found to exhibit good metrics in positive affect, flow, and competence, with low levels of stress and negative affect and a moderate level of challenge. The most critical points raised in the analysis were the need to introduce improvements aimed at enhancing the sensory and imaginative immersion, either through mechanisms to prevent the problem of excessive crowding of groups in the museum space or via refinements to the multimedia elements, giving more importance to visual and audio immersion and de-emphasizing narrative.

Hypothesis H2 stated that non-technical users would be able to use the tool to transform a design script into a playable adventure. The results confirmed this hypothesis, as these users reported positive usability levels for most of the dimensions evaluated. This was not necessarily the case in the context of hypothesis H3, which considered the situation where designers would have to use the tool to create complex designs over an extended period of time and without a reference script. In the latter, general usability metrics, without being too poor, were found to show room for improvement when compared to commonly accepted industry standards.

One of the most prominent of these points is the general desire of designers to feel explicit control over the aesthetics and the visual and acoustic aspects of the system, in line with the findings of the game experience study. Additionally, users expect the system to go beyond providing a platform for the implementation of ideas by offering mechanisms to support actual design from within the tool itself. This observation is consistent with the expectations outlined at the beginning of Section 5.2, where we argued that the concessions made to the authoring system in order to simplify the creative process could prove limiting for users; while we regard the reduction in expressivity as a necessary compromise to ensure the democratization of these authoring platforms to the general public, these insights allow us to identify those points where relaxing the rigidity of these concessions may be desirable. It is also important to note that games of this complexity are usually constructed on the foundation of a proper design document, external to the development engine itself.

In general, the results of the study support the usability of the evaluated tool and user satisfaction with the games derived from it. However, some areas for improvement were highlighted, such as a greater focus on the introduction of design support mechanisms or more freedom of visual and acoustic configuration for creatives. Study participants showed consistent enthusiasm for both the games and the authoring tool, expressing a desire to continue using them to craft new digital experiences in the future. These findings support the idea that our ENIGMACHINE ecosystem has the potential to be an effective and accessible tool for building immersive activities with low development costs in the domain of cultural heritage.

These findings also have implications for other developers interested in producing similar tools and games, who can use this knowledge to devise more engaging and easily adoptable games and authoring systems for non-technical users, as well as for researchers, who can continue to explore the impact and challenges of these authoring pipelines in related institutions. One consideration for future research, however, is the evaluation of the authoring process by domain experts—e.g., museum curators, tourism officers, and so forth—as these latter are closer representations of the intended users of the platform. While having undergraduate students with a game design background (and some programming skills) provided valuable early insights, involving actual domain experts in subsequent evaluations will be crucial to more effectively establish the tool’s suitability and usefulness for its intended users.

In the future, our main goal is to improve the overall satisfaction among users of the tool and the gaming experience of players interacting with the museum applications. All the insights described here will be taken into account in future development iterations, as well as considering the integration of new methodologies such as large language models to further support the creative process, following the trend of other authoring platforms such as Kahoot! or Notion. Meanwhile, and also stemming from the input of the study participants, we contemplate including new phases and interaction systems into the system to enrich the spectrum of options available to the user, especially in terms of increasing variety in the game flow, such as inventory systems, free exploration, branching narratives, etc. By investigating different interaction modalities and editing interfaces, we believe that developers and researchers can continue to innovate and improve the design of interactive content authoring tools.

CRedit authorship contribution statement

Pablo Gutiérrez-Sánchez: Writing – original draft, Validation, Writing – review & editing, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pedro P. Gómez-Martín:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Pedro A. González-Calero:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Marco A. Gómez-Martín:** Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of competing interest

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Data availability

Data will be made available on request.

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